MIL-T-50382(MU)
16 June 1970

MILITARY SPECIFICATION

TELESCOPE, PANORAMIC: M113E1

1. SCOPE

1.1 This specification covers one type of panoramic telescope which provides four-power magnification and a ten-degree field of view. It is equipped with numerical counters which indicate deflection settings and corrections. A resettable counter provides angular readings from a reference point. The reticle pattern and the counter numerals are radioactively illuminated.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments, General Specification Governing the Manufacture, Assembly and Inspection
MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, SUppLy and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures anti Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F11730267	Telescope, Panoramic: M113E1

FSC 1240

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PACKAGING DATA SHEET

11730267 Packaging of Telescope, Panoramic: M113E1

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document fores a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEFENSE SUPPLY AGENCY MANUAL

DSAM 4158.8 Radioactive Commodities in the DOD supply systems

(Copies DSAM 4158.8 may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENT

3.1 Regulatlons pertaining to radioactive devices.- Marking, identification, handling, and shipping of radioactive devices shall be in accordance with DSAM 4158.8.

3.2 Fabrication. - The telescope shall be manufactured in accordance with Drawing F11730267.

3.3 General specification.- The following requirements of MIL-F-13926 shall be met:

a. Order of precedence.
b. Dimensions and tolerances.
c. Inorganic protective surface finishes.
d. Part identification and marking.
e. Workmanship.

3.4 Environmental.

3.4.1 Storage temperatures. - The telescope shall show no evidence of physical failure due to being thermally stabilized at -80° and +160° Fahrenheit (F).

3.4.2 Operating temperatures. - The telescope shall meet the applicable requirements of 3.9 while thermally stabilized at -40° and +150°F.

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3.4.3 Vibration. - The telescope shall be vibrated in a vertical plane at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch total excursion) for a period of five minutes plus or minus fifteen seconds. Subsequent to vibration there shall be no evidence of physical failure.

3.5 Cleanliness. - The telescope shall meet the cleanliness requirements of MIL-O-13830 and there shall be no more than 2 particles of dirt visible on the reticle. The area of each Particle shall not be in excess of the areas equivalent to a size 10 dig. There shall be no dirt on other surfaces which interferes with observation or adversely affects image quality.

3.6 Sealing.

3.6.1 The optical assembly portion. - The optical assembly portion of the telescope shall be capable of sustaining an internal pressure of 5.0 ± 0.1 pounds per square inch gage (psig), with less than 0.2 psig loss, for a minimum period of two hours.

3.6.2 Counterbox assembly portion. - The counterbox assembly portion of the telescope shall be capable of sustaining an internal pressure of 2.0 ± 0.2 psig, with less than 0.2 psig loss, for a minimum period of two hours.

3.7 Accuracy.

3.7.1 Collimation. - With the flat mounting surface of the telescope in a horizontal plane the azimuth counter reading 3200 mils the reset counter shall read 3200 mils when the left (L) and right(R) gunner's aid counters are set at 00 and 00. The eyepiece elbow shall be positioned in detent and shall be perpendicular to the plane of the telescope locating keyways. With the head cover open, the forward line of sight shall be to the left of the eyepiece elbow and shall be parallel to the plane of the telescope locating keyways and horizontal to the seating surface.

3.7.1.1 Open sight. - With the telescope positioned as in 3.7.1, the line of sight of the open sight shall be horizontal within plus or minus 10 mils and parallel in deflection (azimuth) to the optical lines of sight within plus or minus 10 mils.

3.7.1.2 Eyepiece elbow movement. - Movement of the eyepiece elbow from its midpoint to the left and right extremes of travel shall not cause a deviation in excess of one degree.

3.7.2 Image tilt. - The image of a vertical target line shall be vertical within one degree throughout one revolution of the head.

3.7.2.1 Image wander. - The wander of the reticle image measured at the eyepiece shall not exceed 4 degrees in diameter for 2 full revolutions

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of the head.

3.7.3 Plumb travel. - With the telescope positioned as in 3.7.1, the line of sight shall follow a plumb line (starting with the line of sight horizontal and in coincidence with the plumb line) within 0.5 mil when the head prism is depressed a minimum of 300 mils and elevated 300 mils from the horizontal when checked at any deflection setting.

3.7.4 Parallelism of reticle and image. - With the telescope position ed as in 3.7.1, the vertical reticle line shall be parallel to the image of a plumb line within 30 minutes of arc at any deflection setting.

3.7.5 Level travel. - With the telescope positioned as in 3.7.1, the line of sight shall not deviate from a horizontal target line by more than 1.0 mil (total excursion of 2 mils when the head is rotated through. two revolutions.

3.7.5.1 Lift. - Vertical displacement of the line of sight due to reversing direction of rotation of the head shall not exceed 1.0 roil.

3.7.6 Deflection error.

3.7.6.1 Throughout 800 mil steps. - Error at deflection readings in 800 mil steps throughout two full revolutions of the head in one direction shall not exceed 1.0 mil including backlash. The total spread between the greatest positive error and the greatest negative error shall not exceed 1.0 mil. The total error, excluding backlash, between the 6400 mil azimuth counter reading and the reset counter for any one number shall not exceed 0.25 mil.

3.7.6.2 Throughout 15 mil steps. - Error at deflection readings in 15 mil steps taken at eight random positions throughout one full revolution of the head shall not exceed 0.2 mil excluding backlash at each position.

3.7.6.3 Backlash. - Backlash in the deflection mechanism shall not exceed 0.5 mil when read on the 6400 mil azimuth counter and 0.75 mil when read on the reset counter. Backlaxh in the elevating mechanism shall not exceed 1.0 mil.

3.7.7 Eyepiece focus. - Eyepiece focus of the horizontal reticle line and vertical reticle line at the center of the field shall each be within the limits of 0.75 and 1.0 diopters.

3.7.8 Resolution. - Resolution on the optical axis shall be 15 seconds of arc or less when using an auxiliary telescope of at least 3 power having an entrance pupil of at least 0.4 inch. Resolution shall consist of detectable line structure and proper line count in all 4 meridians. Variation in focus between meridians shall not exceed 0.25 diopters as measured at the panoramic eyepiece.

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3.7.9 Parallax. - Parallax in the center of the field shall not exceed 0.35 mil when viewing a target between 40 meters and infinity. (Parallax is nominally zero at 80 meters.)

3.7.10 Reticle accuracy. - The total angular subtense of the reticle graduations in object space shall be as indicated on the reticle drawing within two percent.

3.8 Counter operation.

3.8.1 Reset counter. - The reset counter shall be resettable to 3200 mils from any counter reading. The reset counter and the 6400 mil azimuth counter shall follow when the operating knob is turned at the rate of 200 revolutions per minute (RPM) for a period of one minute. The gunner's aid counter shall remain at R-00 and L-00.

3.8.2 Gunner's aid counter. - The stops shall allow an excursion of plus 50 mils and minus 50 mils

3.8.2.1 Counter setting. - Setting right (R) 50 mils on the gunner’s aid counter shall increase the reset counter by 50 mils. Setting left (L) 50 mils on the gunner’s aid counter shall decrease the reset counter by SO mils No change shall result in the 6400 mil azimuth counter.

3.8.2.2 Direct fire, click lead, engagement knob. - With the detent on the knob of the 6400 mil azimuth counter Set for direct fire the azimuth counter knob shall indicate a distinguishable click, indicating detent position at each five mil internal between 0 and 6400 mils

3.9 Operability of movable Parts.

3.9.1 Torque. -The running torque required to rotate the following components shall be within the values tabulated. When checked at the temperature specified in 3.4.2, the running torque values shall not exceed the value obtained at standard ambient temperature by more than 150%.

Elbow Assembly	not to exceed 12 inch pounds
Elevation Knob	5 to 48 inch ounces
Reset Counter (rotation to reset counter)	2 to 40 inch ounces
Azimuth Knob	4 to 10 inch pounds
Boresight Adjustment (rotation while clutch is disengaged)	2 to 20 inch ounces

3.9.2 Operability. - This requirement shall be met with the telescope thermally stabilized at the temperatures specified in 3.4.2. The reset counter knob shall be manually operable to push in for resetting counters (direct force). The boresight adjustment shall be operable to disengage the clutch for boresighting (direct force). The azimuth counter cover

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and latch, head cover and latch, azimuth knob folding handle, elbow detent lock pin and spring action of the electrical contact shall be operable.

3.10 Illumination.

3.10.1 Reticle. - This requirement shall be met with the line of sight parallel flat mounting surface within 10 mils. The reticle markings shall be visible in ambient light conditions ranging from dusk into night. When required, the use of a mask with a 1/2" aperture, centrally located on the entrance window, will be permitt to meet this requirement.

3.10.2 Counters. - The counter numerals shall be visible through all ambient light conditions.

3.11 Counter numerals. - With the lowest digit of each counter in coincidence with its respective index line, the remaining numbers shall be in horizontal alignment within 1/16".

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract of purchyase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. - The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial Production) approval. - The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) telescopes plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same reamer, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. - When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing. - When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection Provisions.

4.3.1 Submission of product. - Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in-accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. - All components and sub-assemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAPS, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. - Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (Individual characteristic) basis in accord-ance with MIL-STD-105 and the sampling plans specified in Table I herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classifi-cation of defects in Table I shall constitute the miniuum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	CHARACTERISTIC		TEST PROCEDURE
CRITICAL:	100% Inspection		
1.	Pressure (optical assembly)	3.6.1	4.6.1.1
2.	Pressure (counter assembly)	3.6.2	4.6.1.1.1
MAJOR :	AQL 1.0% Defective		
101.	Previbration collimation	3.7.1	4.7.2.1
102.	Vibration	3.4.3	4.7.2.1.1
103.	Post-vibration collimation	3.4.3	4.7.2.1.2
104.	Image wander	3.7.2.1	4.7.2.1.3
105.	Cleanliness	3.5	4.7.2.1.4
106.	Image tilt	3.7.2	4.7.2.1.5
107.	Parallelism of image & reticle	3.7.4	4.7.2.1.6
108.	Eyepiece focus	3.7.7	4.7.2.1.7
109.	Parallax	3.7.9	4.7.2.1.8

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TABLE I - Cont'd

110.	Reticle accuracy	3.7.10	4.7.2.1.9
111.	Settings, counter, gunner's aid	3.8.2.1	4.7.2.2.1
112.	Counter stops	3.8.2	4.7.2.2.1
113.	Deflection error (800 mil)	3.7.6,1	4.7.2.2.2
114.	Deflection error (15 mil)	3.7.6.2	4.7.2.2.2
115.	Azimuth backlash	3.7.6.3	4.7.2.2.2
116.	Level travel	3.7.5	4.7.2.2.3
117.	Lift	3.7.5.1	4.7.2.2.4
118.	Click mechanish	3.8.2.2	4.7.2.2.5
119.	Rapid traverse (reset counter)	3.8.1	4.7.2.2.6
120.	Resolution	3.7.8	4.7.2.2.7
121.	Plumb travel	3.7.3	4.7.2.3.1
122.	Torque (60°-90°F)	3.9.1	4.7.2.4.1
123.	Operability (60°-90°F)	3.9.2	4.7.2.4.1
124.	Illumination (reticle)	3.10.1	4.7.2.1
125.	Illumination (counters)	3.10.2	4.7.2.2
126.	Elbow movement	3.7.1.4	4.7.2.1.6
127.	Counter numbers	3.11	4.7.2.6

MINOR: AQL 2.5% Defective

201.	Head cover	3.7.1.1	4.7.2.1.2
202.	Open sight	3.7.1.3	4.7.2.2.8
203.	Excursion range (elevation)	3.7.3	4.7,2.3.1
204.	Elevation backlash	3.7.6.3	4.7.2.3.2

4.3.3 Acceptance and rejection. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling for environmental testing.

4.4.1 Samples shall be selected from lots at random using inspection Level S-3 Table I of MIL-STD-105. Accptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of MIL-STD-105. The samples shall have met all the requirements l nd tests outlined in Table I prior to being subjected to the following examination and test.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Regulation pertaining to radio-active devices	3.1	Certification
302.	Fabrication	3.2	Certification
303.	General specification	3.3	Certification
304.	Temperature storage	3.4.1	4.6.1.2
305.	Temperature operating	3.4.2	4.6.1.2
306.	Operability (storage & oper. temp)	3.9.2	4.7.2.4
307.	Torque (storage & operating tamp)	3.9.1	4.7.2.4.1

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4.4.2 Failure of sample. - Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design. - Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design. - The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this Inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90%. of the prescribed tolarance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment. - In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6 and 4.7.

NOMENCLATURE	DESCRIPTION
1. Hot and cold chamber	Standard type conforming to the accuracies outlined in test facilities specified under test facilities of MIL-F-13926

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2.	Power supply	0 to 1000VDC output within 10% accuracy
3.	Torque wrench and scale	Standard shop type accurate to within 10% of drawing tolerance

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Temperature test. - The panoramic telescope shall be subjected to test Procedure I of MIL-F-13926; however, the temperatures specified in 3.4.1 and 3.4.2 shall apply. When the panoramic telescope has been stabilized at the temperature specified, the result of the specified test shall conform to the limits outlined in 3.4.1 and 3.4.2 using test procedures outlined herein.

4.6.1.2 Sealing optical assembly.- This test shall be performed using testing equipment the accuracy of which shall conform to the accuracy dipicted on Drawing F856556. When using pressure test fixture, the fixture shall be set-up as-outlined in set-up instructions shown on Drawing F8565556, Sheet 1. The gas used shall be dry nitrogen having a dew point of -25°F. After connecting the test fixture to the panoramic telescope, adjust testing pressure to the pressure specified in 3.6.1 l nd maintain the pressure for time specified in 3.6.1. Pressure gage shall remain constant to assure compliance with 3.6.1. At the completion of this test, connect a dew point tester to the exit post and purge the optical assembly until -25°F is recorded and seal the unit.

4.6.1.2.1 Sealing (counter assembly). - This test to be accomplished as outlined above, however, the pressure and time values specified in 3.6.2 shall apply.

4.7 Performance testing.

4.7.1 Orientation. - Proper orientation of the panoramic telescope is assured when mounted and secured to an adapter, the accuracy of which shall be equal to or better that the accuracy depicted on Drawing F8565945 and when aligned as outlined in set-up instuctions spectfied on Drawing F8213899. With the panoramic telescope positioned on the fixture as outlined above the line of sight, elbow assembly, counters and the five mil click mechanism shall be set as specified in 3.7.1.

4.7.2 Collimation.

4.7.2.1 Previbration. - With telescope positioned as outlined in 4.7.1, observe coincidence between the panoramic telescope line of sight and the optical axis of the test reticle (target) in the collimator pro-

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jector. This coincidence must be within the limits outlined in 3.7.1. Record this position. Exercise care not to move the setting of the panoramic telescope.

4.7.2.1.1 Vibration. - This test shall be performed using a testing device, the accuracy of which shall be equal to or exceed the accuracy depicted on Fixture Drawing F7560085. The test shall be perfomed as outlined in Procedure V of MIL-F-13926; however, the duration shall be as specified in 3.4.3. At the conclusion of the test, the telescope shall be subjected to a visual and tactile examination and shall be subjected to the following performance test to determine conformance with 3.4.3.

4.7.2.1.2 Post vibration. - Immediately after completing the vibration test, reassemble the telescope to the adapter and without disturbing any previous settings, observe where the panoramic telescope's optical line of sight falls upon the projector collimator reticle. This observed point and the recorded position (4.7.2.1) shall coincide within the limits specified in 3.4.3. Close the cover and observe where the line of sight falls with respect to the recorded position of the optical axis. They must coincide within the limits outlined in 3.4.3.

4.7.2.1.3 Image wander. - With the cover opened and with the telescope positioned as outlined in 4.7.1, rotate the telescope azimuth mechanism in one direction to any azimuth setting and rotate the azimuth mechanism of the test fixture an equal amount in the opposite direction. Now, observe coincidence between the optical axes. Non-coincidence between the axes shall not exceed an 18 mil spread to assure compliance with 3.7.2.1. This spread to be measured on the panoramic telescope reticle.

4.7.2.1.4 Cleanliness. - The panoramic telescope shall be tested for cleanliness by using the shadowing technique outlined in MIL-O-13830. At the conclusion of this test, dirt particles shall not exceed the limits outlined in 3.5.

4.7.2.1.5 Image tilt. - With the panoramic telescope and fixture positioned as outlined in 4.7.1, image tilt shall be observed by sighting through Collimator F5549108 aligned to the optical axis of the eyepiece assembly and note parallelism between its vertical reticle line and the vertical line of the collimator target. Both lines shall be parallel within the limits provided on the projected target (collimator reticle) to assure conformance with 3.7.2. Repeat this test with the elbow assembly against each of the stops to assure conformance with 3.7.1.4. Again observe parallelism between the reticle lines. Repeat this test at eight different azimuth counter settings in 6400 mils of rotation. Each setting in azimuth will require the azimuth fixture plate to be rotated an equal amount in an opposite direction to permit reestablishing the line of sight to the position observed in 4.7.2.1.2. All readings are to be within the limits outlined in 3.7.1.4.

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4.7.2.1.6 Parallelism of image and reticle. - With the panoramic telescope positioned as outlined in 4.7.1, align one end of the instrument's vertical reticle line to the projector collimator's vertical reticle line and observe where the opposite end of the panoramic telescope vertical reticle line falls with respect to the projector collimator vertical line, both lines must coincide within the limits specified in 3.7.4. Repeat this procedure with elbow assembly against each of the extreme stops. Also repeat this procedure at approximately eight different azimuth settings 6400 mils of rotation. Return the line of sight to the projector collimator's target line after each selected azimuth setting by rotating the azimuth fixture plate an equal amount in the opposite direction. All check points are to be within the limits outlined in 3.7.1.4 and 3.7.4.

4.7.2.1.7 Eyepiece focus. - This test is to be accomplished by placing a white background in front of the panoramic telescope window. Eyepiece focus is to be measured using a 3 power dioptometer conforming to the accuracies depicted on Drawing 7680631. Adjust the dioptometer eyepiece assembly to the dioptometer's reticle until a clear sharp image appears on the dioptometer's optical axis. Nest direct the diptometer's line of sight toward a 500 feet distant target such as a smoke stack, edge of a building, or some other well defined target image. Adjust the objective assembly until parallax is removed. Parallax is defined as any apparent movement between the reticle and target image when the head is moved from side to side or up and down. Align the dioptometer's optical axis to the panoramic telescope optical axis and obtain a clear sharp image of the panoramic telescope reticle on the optical axis. Make all measurements on the dioptometer by adjusting the objective assembly. Any and all adjustments to be within the limits of 3.7.7.

4.7.2.1.8 Parallax. - Using the set-up outlined in 4.7.1, parallax will be checked after each of the distances specified in 3.7.9 has been set into the projector collimator. The test is performed by aligning a reticle line or mark near or on the optical axis of the projector collimator reticle. Observe this positioning, there shall be no apparent movement between the selected lines or mark when the observer moves his head from side to side or up and down. Reticle displacement shall not exceed the limits outlined in 3.7.9. Use the thickness of the reticle line to measure apparent movement.

4.7.2.1.9 Reticle accuracy. - Reticle accuracy may be assured by aligning the intersecting point of the panoramic telescope's vertical and horizontal reticle line to the corresponding intersecting reticle lines of the projector collimator. Adjustments may be made by means of the panoramic telescope's azimuth and elevating mechanisms. When coincidence has been established, observe where the 80 mil graduations of the panoramic telescope's horizontal reticle line falls with respect to the corresponding reticle line of the projector collimator. They must coincide within the limits of the reticle tolerance mark to assure compliance with 3.7.10.

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4.7.2.2 Counter operation.

4.7.2.2.1 Gunner aid counters. - This test may be accomplished with the panoramic telescope positioned as outlined in 4.7.1 with the line of sight falling upon the projector collimators' reticle as outlined in 4.7.2.1.2. Observe and record the azimuth test fixture reading. Set any correction reading into either the "L" or "R" counter. Any reading induced Into the right "R" gunner's aid counter shall increase the reset counter an equal amount and any reading induced into the left "L" gunner's aid counter shall decrease the reset counter an equal amount to assure compliance with 3.8.2.1. When 50 mils has been inserted into either gunner's aid counter, the reset counter shall reflect an increase or decrease of 50 mils within the limits outlined in 3.8.2.1 and the correction counter shall have traveled the minimum excursion outlined in 3.8.2 and the 6400 mil counter shall remain unchanged (see 3.8.2.1). The test for accuracy shall be performed by inserting a correction into either the "L" or "R" gunner's aid counter and observing the appropriate change in the reset counter. Turn the azimuth fixture plate to. traverse the line of sight in a clockwise direction an equal amount when a correction is placed into the "L" counter By means of the azimuth knob of the panoramic telescope, traverse the line of sight in the opposite direction until the reset counter records 3200 mils. Now sight through the panoramic telescope and observe coincidence between the panoramic telescope's optical axis and the point on the projector collimator's reticle established in 4.7.2.1.2. This coincidence must be maintained with the limits specified in 3.8.2.1. Repeat this procedure at two additional correction readings and at the 50 mil correction setting of each gunner's aid counter.

4.7.2.2.2 Deflection error, backlash and 15 mil settings. With the panoramic telescope positioned as outlined In 4.7.1, rotate the azimuth fixture plate 800 mils. Sight through the panoramic telescope and rotate the azimuth knob in the opposite direction until the line of sight falls in coincidence with the position outlined in 4.7.2.1.2. Do not override this point of coincidence. Record deflection counter reading and reset counter reading. Both should be in conformance with 3.7.6.1. Rotate the deflection knob one complete revoltution and return the line of sight to the previously established position and observe deflection and reset counter readings. This difference, if any, is backlash and shall not exceed the limits of 3.7.6.3. Rotate the azimuth plate an additional 15 mils and repeat above procedure. When coincidence is obtained observe deflection and reset counter readings. Coincidence must conform to the limits outlined in 3.6.7.2. Repeat the 15 mil check after each 800 mil setting.

4.7.2.2.3 Level travel. - This test may be performed in conjunction with the deflection test outlined in 4.7.2.2.2. The test is performed by sighting through the panoramic telescope after a selected azimuth setting has been inserted into the fixture and panoramic telescope, observe coincidence between the telescope horizontal reticle line and the horizontal target line near the optical axis. They must coincide within the limits outlined in 3.7.5.

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4.7.2.2.4 Lift. - This test may be performed In conjunction with the backlaxh test outlined in 4.7.2.2.2. The test is performed after returning the line of sight to the point of coincidence by observing where the horizontal reticle line of the panoramic telescope falls with respect to the previously established position (4.7.2.2.3). They must coincide within the limits of 3.705.1.

4.7.2.2.5 Five mil click mechanism. - This test may be performed by using visual and tactile inspection techniques. The result of this inspection shall conform to the requirement outlined in 3.8.2.7 when performed at the temperature specified.

4.7.2.2.6 Rapid traverse (reset counter). - This test is made by assembling a testing mechanism capable of providing an RPM output specified in 3.8.1 to the azimuth knob of the panoramic telescope by means of-an appropriate adapter. After subjecting the panoramic telescope to the RPM for the time specified, the reset counter and the azimuth counter shall follow as required in 3.8.1 and the reset counter shall reset to 3200 mils at the end of each speed test required in 3.8.1. The gunner’s aid counter shall remain as specified in 3.8.1.

4.7.2.2.7 Resolution. - This test may be performed after directing the line of sight toward a target similar to that specified in MIL-O-13830. The actual test is performed as outlined in MIL-O-13830. Observe the definition pattern on the target. The patterns must be resolved in the four meridians within the limits outlined in 3.7.8.

4.7.2.2.8 Open sight. - This test is accomplished by using the setup in 4.7.1. The test is performed by positioning the optical axis of the auxiliary telescope, F5549108 close to, but not on the vertex of the “V” grooves of the open sight and observing where the optical axis of the auxiliary telescopes fall upon the target with respect to the 20 mil square of the target, see Sheet 8 of Drawing F8213899. The optical axis umst fall within the square to conform with the tolerance specified in 3.7.1.1.

4.7.2.3 Elevation.

4.7.2.3.1 Elevation and plumb travel. - Return the panoramic telescope to the position outlined in 4.7.1. Scribe an index line on the elevation knob in line with an index scribed on the head. Perform these tests-by. rotating the azimuth fixture plate to direct the panoramic telescope’s optical axis toward a plumb line. Position a plumb line in coincidence with the center vertical reticle line. Position the vertical target line in coincidence with the plumb and vertical reticle line and raise or lower the target to obtain coincidence between the target horizontal line and the horizontal reticle line of the panoramic telescope’s reticle. The plumb travel test is performed by actuating the panoramic telescope’s elevation knob from the zero setting to each extreme. The line of sight must travel to the limit lines of the target to assure compliance with 3.7.3 and must follow the center vertical line from zero elevation within the width of the target limit line to assure compliance with 3.7.3. Repeat this procedure at approximately six different azimuth settings by turning the panoramic tele-

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scopes's azimuth mechanism in one direction and toe rotating surface of the test fixture in the opposite direction to obtain the point of coincidence mentioned above.

4.7.2.3.2 Elevation backlash. - This test can be performed in conjunction with the previous test by sighting through the panoramic telescope and by means of the elevation knob raise or lower the line of sight to a selected line on the target (do not override this target line) and scribe a line on the head casting in line with the index on the elevation knob. Rotate the elevation knob past this selected target line and return to the coincided lines mentioned and observe where the optical axis (line of sight) falls with respect to the previously selected target line. This This difference, if any, as measured by the thickness of the reticle line, shall not exceed the limits outlined in 3.7.6.3.

4.7.2.4 Operability. - When the panoramic telescope has been stabilized at the extreme operating temperatures specified in 3.9.2, tactile inspection techniques shall be used to determine compliance outlined in 3.9.2.

4.7.2.4.1 Torque. - Torque tests shall be performed under temperature conditions outlined in 3.9.1. The torque tests shall be performed using a torque wrench and appropriate adapter. The torque readings obtained shall conform to the limits specified in 3.9.1.

4.7.2.5 Illumination.

4.7.2.5.1 Reticle illumination. - Place the instrument in a dark room having a controlled light source. The observer and instrument must become dark adapted. Prior to performing the test, place a white matted surface approximately 2 feet from the telescope window. Perform the test by observing reticle pattern and determine that this pattern is clearly discernible by means of the self-illuminating source in the dark room. Gradually raise the level of illumination to where the pattern becomes indiscernable. Place a fixed stop having an aperture as specified in 3.10.1 over the objective window and observe that the reticle pattern is again clearly discernible by means of the self-illuminating source. Raise ambient the level of illumination to 10 foot candle, remove the fixed stop and observe the teticle pattern. It shall appear as when viewed in ambient light condition to assure conformance with 3.10.1.

4.7.2.5.2 Counter illumination. - Place the instrument in a dark room having a controlled light source. The observer and instrument must become dark adapter. After being adapted to the darkness, the observer shall visually examine the counter numbers and lines. They must be clearly discernible to assure conformance with 3.10.2. Gradually raise the ambient level of light in the dark room 1 foot candle of illumination in the area of the counter. The counter numbers and lines shall be clearly discernible to assure conformance with 3.10.2.

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4.7.2.6 Counter numbers. - By means of standard measuring equipment, determine digit alignment in accordance with 3.11.

5. PREPARATION AND DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Packaging Data Sheet 11730267. The level is protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Panoramic Telescope: M113E1 is primarily intended to lay artillery weapons in indirect fire. This telescope can also be used for direct fire in azimuth.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Selection of applicable levels of preservation, packaging and packing
c. Applicable stock number
d. Revisions for First Article testing

6.3 Radioactive material. - Respective bidders should be cognizant of the need for an Atomic Energy Commission license governing the procurement and handling of radioactive material.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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